Department of Electrical & Software Engineering
University of Calgary
ENSF 337 - Programming Fundamentals
Term Project
Fall 2026

Objectives:
The objective of this term project is to give you an opportunity to apply the principles of object-oriented software development in an application that is relatively close to a real-world example, but still on a much smaller scale and without many real-world software features such as client-server and web-based systems.

Total Mark: 100
This is a team-based assignment. **Each team may have no more than three students**. All team members are expected to participate in the design, implementation and testing of the program. Each team must submit a separate contribution summary.

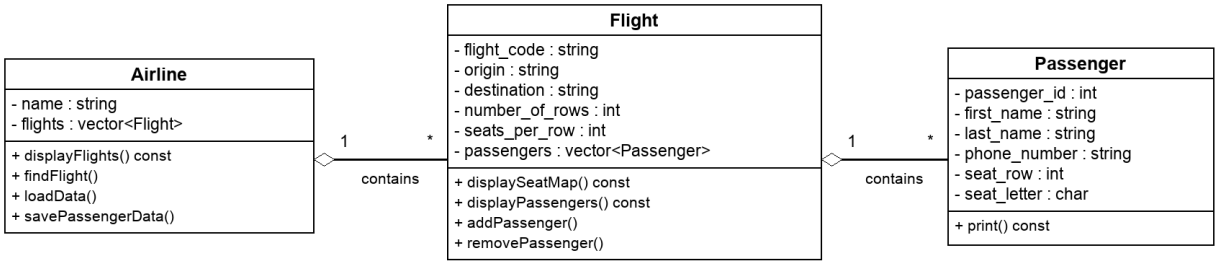
Due Date:
The project is due on November 29th, 2026.

Presentation Dates:
Project presentations will take place between November 30th and December 4th, 2026.

What to Submit:
1. One PDF report with the team members' names and UCIDs, a brief explanation of the class design and program data flow, and representative test cases. Each test must state the input or action, expected result, actual result, and whether the test passed. The tests must use both input-file sets.
2. One separate PDF contribution summary that states what each team member designed, implemented and tested.
3. One ZIP file containing all .h and .cpp files, the two supplied data files, the two team-created test data files, and a README.txt with the exact compile and run commands for Windows and macOS. Do not include an .exe, a.out, or other compiled files.

Project Description:
In this project you are going to develop a small flight-management application whose overall functionality includes:
- Loading flight and passenger data from two text files
- Allowing the user to browse and select a flight
- Allowing the user to display a flight seat map and passenger list
- Allowing the user to add a passenger to a flight
- Allowing the user to remove a passenger from a flight
- Allowing the user to save the updated passenger data back to the file

Minimum Requirements of your Project:
Implement the project using the following class structure. You may add private helper functions, constructors and accessors where needed, but you must preserve these three main classes and their relationships.



How the Application Is Supposed to Work:

Project Description:

- The program starts with a title as follows:

```
FMAS Version: 1.0
Term Project - Flight Management Application System
Produced by group: Your group number
Names: Your team members' names

<<< Press Return to Continue>>>>
```

- When the user presses Return, the program enters a menu-driven interaction mode that allows the user to:
  - display all flights and select one flight
  - display the selected flight's seat map and passenger list
  - add a passenger to the selected flight
  - remove a passenger from the selected flight
  - save the passenger data when requested
  - terminate the program
- The application starts by reading the two supplied files that contain existing data:

1. A text file that contains flight information, called flights.txt:

```
AC210   Calgary    Vancouver   8    4
WS318   Calgary    Toronto    10    4
PD427   Edmonton   Victoria    7    4
F8205   Calgary    Montreal    9    4
```

The columns are flight code, departure city, destination city, number of seat rows, and number of seats per row. For this project, city names contain no spaces. Seat rows are numbered from 1 through the stated number of rows.

2. A text file that contains passenger information, called passengers.txt:

```
AC210   26001   Maya   Chen     403-555-0114   2   B
AC210   26002   Noah   Patel    403-555-0187   5   D
WS318   26003   Liam   Brooks   587-555-0132   1   A
PD427   26004   Ava    Singh    780-555-0166   3   C
```

The columns are flight code, passenger ID, first name, last name, phone number, seat row, and seat letter. You may assume that names and phone numbers contain no spaces.

The supplied project package contains flights.txt and passengers.txt with the data shown above. Do not hard-code these records in the program.

Each team must also create flights_test2.txt and passengers_test2.txt as a second input set. It must use different values, contain at least two valid flights and at least four valid passengers across at least two flights, and contain no duplicate passenger IDs or duplicate occupied seats on the same flight. Include both files in the ZIP and show tests using them in the report.

Implementation Requirements:

- Use the Passenger, Flight and Airline classes shown on page 1. Airline stores a vector<Flight>, and each Flight stores a vector<Passenger>.
- Build the seat map for display using vector<vector<char>>. Mark an available seat with '-' and an occupied seat with 'X'.
- The supplied FileRecordIO.h and FileRecordIO.cpp contain the C++ stream code that reads and writes the record files. Use these functions from Airline::loadData and Airline::savePassengerData. You are responsible for converting the loaded records into Flight and Passenger objects and for assembling the passenger records that are saved.
- The program must work with either input-file set by changing only the filenames passed to the loading functions. Do not hard-code the supplied flights, passengers or seat map.

The next few pages show a sample run of the expected program. The exact spacing and wording may vary, but the required information and behaviour must be present.

Note: the text in RED is the user input

```
Version: 1.0
Term Project - Flight Management Program in C++
Produced by: Your team names
Year: 2026

<<< Press Return to Continue>>>>

Please select one of the following options:

   1. Display all flights.
   2. Select a flight.
   3. Display Flight Seat Map.
   4. Display Passengers Information.
   5. Add a New Passenger.
   6. Remove an Existing Passenger.
   7. Save data.
   8. Quit.

Enter your choice: (1, 2, 3, 4, 5, 6, 7, or 8) 1

Here is the list of available flights:

   1. AC210   Calgary    Vancouver    8   4
   2. WS318   Calgary    Toronto      10   4
   3. PD427   Edmonton   Victoria      7   4
   4. F8205   Calgary    Montreal      9   4

<<< Press Return to Continue>>>>

Please select one of the following options:

   1. Display all flights.
   2. Select a flight.
   3. Display Flight Seat Map.
   4. Display Passengers Information.
   5. Add a New Passenger.
   6. Remove an Existing Passenger.
   7. Save data.
   8. Quit.

Enter your choice: (1, 2, 3, 4, 5, 6, 7, or 8) 2

Enter the flight number: 1
You have selected flight AC210 from Calgary to Vancouver.

<<< Press Return to Continue>>>>

Please select one of the following options:

   1. Display all flights.
   2. Select a flight.
   3. Display Flight Seat Map.
   4. Display Passengers Information.
   5. Add a New Passenger.
   6. Remove an Existing Passenger.
   7. Save data.
   8. Quit.

Enter your choice: (1, 2, 3, 4, 5, 6, 7, or 8) 3

Aircraft Seat Map
      A   B   C   D
 1   -   -   -   -
 2   -   X   -   -
 3   -   -   -   -
 4   -   -   -   -
 5   -   -   -   X
 6   -   -   -   -
 7   -   -   -   -
 8   -   -   -   -

<<< Press Return to Continue>>>>
```

```
Please select one of the following options:

   1. Display all flights.
   2. Select a flight.
   3. Display Flight Seat Map.
   4. Display Passengers Information.
   5. Add a New Passenger.
   6. Remove an Existing Passenger.
   7. Save data.
   8. Quit.

Enter your choice: (1, 2, 3, 4, 5, 6, 7, or 8) 4

Passenger List (Flight: AC210 from Calgary to Vancouver)

ID      First Name   Last Name    Phone           Row   Seat
-------------------------------------------------------------
26001   Maya         Chen         403-555-0114      2     B
26002   Noah         Patel        403-555-0187      5     D
-------------------------------------------------------------
<<< Press Return to Continue>>>>

Please select one of the following options:

   1. Display all flights.
   2. Select a flight.
   3. Display Flight Seat Map.
   4. Display Passengers Information.
   5. Add a New Passenger.
   6. Remove an Existing Passenger.
   7. Save data.
   8. Quit.

Enter your choice: (1, 2, 3, 4, 5, 6, 7, or 8) 5

Please enter the passenger id: 26005
Please enter the passenger first name: Olivia
Please enter the passenger last name: Martin
Please enter the passenger phone number: 403-555-0148
Enter the passenger's desired row: 3
Enter the passenger's desired seat: C
Passenger Olivia Martin was added to flight AC210.

Please select one of the following options:

   1. Display all flights.
   2. Select a flight.
   3. Display Flight Seat Map.
   4. Display Passengers Information.
   5. Add a New Passenger.
   6. Remove an Existing Passenger.
   7. Save data.
   8. Quit.

Enter your choice: (1, 2, 3, 4, 5, 6, 7, or 8) 5

Please enter the passenger id: 26006
Please enter the passenger first name: Elias
Please enter the passenger last name: Romero
Please enter the passenger phone number: 403-555-0199
Enter the passenger's desired row: 7
Enter the passenger's desired seat: A
Passenger Elias Romero was added to flight AC210.

Please select one of the following options:

   1. Display all flights.
   2. Select a flight.
   3. Display Flight Seat Map.
   4. Display Passengers Information.
   5. Add a New Passenger.
   6. Remove an Existing Passenger.
   7. Save data.
   8. Quit.

Enter your choice: (1, 2, 3, 4, 5, 6, 7, or 8) 4

Passenger List (Flight: AC210 from Calgary to Vancouver)

ID      First Name   Last Name    Phone           Row   Seat
-------------------------------------------------------------
26001   Maya         Chen         403-555-0114      2     B
26002   Noah         Patel        403-555-0187      5     D
```

```
26005   Olivia       Martin       403-555-0148      3     C
26006   Elias        Romero       403-555-0199      7     A
------------------------------------------------------------
<<< Press Return to Continue>>>>

Please select one of the following options:

   1. Display all flights.
   2. Select a flight.
   3. Display Flight Seat Map.
   4. Display Passengers Information.
   5. Add a New Passenger.
   6. Remove an Existing Passenger.
   7. Save data.
   8. Quit.

Enter your choice: (1, 2, 3, 4, 5, 6, 7, or 8) 6

Please enter the id of the passenger that needs to be removed: 26005

Passenger Olivia Martin was successfully removed from flight AC210.
<<< Press Return to Continue>>>>

Please select one of the following options:

   1. Display all flights.
   2. Select a flight.
   3. Display Flight Seat Map.
   4. Display Passengers Information.
   5. Add a New Passenger.
   6. Remove an Existing Passenger.
   7. Save data.
   8. Quit.

Enter your choice: (1, 2, 3, 4, 5, 6, 7, or 8) 4

Passenger List (Flight: AC210 from Calgary to Vancouver)

ID      First Name   Last Name    Phone           Row   Seat
------------------------------------------------------------
26001   Maya         Chen         403-555-0114      2     B
26002   Noah         Patel        403-555-0187      5     D
26006   Elias        Romero       403-555-0199      7     A
------------------------------------------------------------
<<< Press Return to Continue>>>>

Please select one of the following options:

   1. Display all flights.
   2. Select a flight.
   3. Display Flight Seat Map.
   4. Display Passengers Information.
   5. Add a New Passenger.
   6. Remove an Existing Passenger.
   7. Save data.
   8. Quit.

Enter your choice: (1, 2, 3, 4, 5, 6, 7, or 8) 7

Save the updated passenger data to passengers.txt? (Y or N) Y

All passenger data was saved successfully.
<<< Press Return to Continue>>>>

Please select one of the following options:

   1. Display all flights.
   2. Select a flight.
   3. Display Flight Seat Map.
   4. Display Passengers Information.
   5. Add a New Passenger.
   6. Remove an Existing Passenger.
   7. Save data.
   8. Quit.

Enter your choice: (1, 2, 3, 4, 5, 6, 7, or 8) 3
```

```
Aircraft Seat Map
      A   B   C   D
 1    -   -   -   -
 2    -   X   -   -
 3    -   -   -   -
 4    -   -   -   -
 5    -   -   -   X
 6    -   -   -   -
 7    X   -   -   -
 8    -   -   -   -

<<< Press Return to Continue>>>>

Please select one of the following options:

   1. Display all flights.
   2. Select a flight.
   3. Display Flight Seat Map.
   4. Display Passengers Information.
   5. Add a New Passenger.
   6. Remove an Existing Passenger.
   7. Save data.
   8. Quit.

Enter your choice: (1, 2, 3, 4, 5, 6, 7, or 8) 8

Program terminated.
```